

A Block-Clustered Generator with a Bounded Non-Riesz Basis of Eigenvectors

Candidate full solution packet for arXiv:1405.2731

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Status. Candidate full solution, likely valid, pending human review. The construction gives an affirmative answer and also produces simple purely imaginary eigenvalues and a linearly bounded group.

1 Source question

Sklyar and Marchenko ask in the concluding remarks of [1, p. 32]:

Is it possible to construct the unbounded generator of a C_0 -group with bounded non-Riesz basis family of eigenvectors?

Here “bounded non-Riesz basis” means a Schauder basis whose vector norms are bounded above and away from zero, but which is not a Riesz basis; see also the authors’ later formulation in arXiv:1809.03079. The source passage is shown below.

Finally, the answer to the following important question does not yet exist. Is it possible to construct the unbounded generator of a C_0 -group with bounded non-Riesz basis family of eigenvectors? We note that the construction of unbounded generator of a C_0 -semigroup with bounded non-Riesz basis family of eigenvectors is quite simple, see, e.g., [13] (Section 5.5).

Figure 1: The complete question and the semigroup comparison on page 32 of the source paper.

2 Idea of the proof

Start from one normalized conditional Schauder basis in a Hilbert space and look at longer and longer finite prefixes. Their basis constants remain uniform, but their Hilbertian condition numbers become arbitrarily large. Place copies of these prefixes in mutually orthogonal finite-dimensional blocks. The concatenation is still a normalized Schauder basis, while its block condition numbers force it not to be Riesz.

An arbitrary diagonal phase multiplier can be large on an ill-conditioned block. The key is to give the vectors in that block almost the same frequency. The common phase is an isometry and disappears from the norm estimate; only the within-block frequency diameter is amplified by the bad condition number. Choosing that diameter inversely proportional to the condition number

yields a uniform group bound. Meanwhile the common frequencies of successive blocks tend to infinity, so the generator is unbounded. This simultaneously reconciles strong continuity, unbounded spectrum, and conditional basis geometry.

3 The result

Theorem 1. *There exist a separable complex Hilbert space H , a strongly continuous group $(T(t))_{t \in \mathbb{R}}$ on H , and its unbounded generator A such that:*

- (i) *A has pairwise distinct purely imaginary eigenvalues;*
- (ii) *corresponding normalized eigenvectors form a Schauder basis of H but not a Riesz basis; and*
- (iii)

$$\|T(t)\| \leq 1 + \frac{|t|}{2} \quad (t \in \mathbb{R}).$$

In particular, the answer to the source question is affirmative.

4 Proof

The existence of a normalized conditional Schauder basis in a separable Hilbert space is classical. For example, the source paper [1, p. 7] recalls Babenko's explicit bounded non-Riesz basis in $L_2(-\pi, \pi)$ [2]. Fix such a basis $(u_j)_{j \geq 1}$ in a Hilbert space H_0 , and let

$$K = \sup_{r \geq 1} \left\| \sum_{j=1}^r u_j^* \otimes u_j \right\| < \infty$$

be its Schauder basis constant.

For $N \geq 1$, put $E_N = \text{span}\{u_1, \dots, u_N\}$ and let

$$S_N : \ell_2^N \longrightarrow E_N, \quad S_N e_j = u_j, \quad \kappa_N = \|S_N\| \|S_N^{-1}\|.$$

Lemma 1. *The sequence (κ_N) is unbounded.*

Proof. Normalization gives $\|S_N\| \geq 1$ and $\|S_N^{-1}\| \geq 1$. If $\sup_N \kappa_N < \infty$, both factors would be uniformly bounded. Hence a constant C would satisfy

$$C^{-1} \left(\sum_{j=1}^N |c_j|^2 \right)^{1/2} \leq \left\| \sum_{j=1}^N c_j u_j \right\| \leq C \left(\sum_{j=1}^N |c_j|^2 \right)^{1/2}$$

for every N and every scalar family $(c_j)_{j=1}^N$. These are precisely uniform Riesz inequalities, contradicting that (u_j) is non-Riesz. $\square$

Choose strictly increasing N_m such that $\kappa_m := \kappa_{N_m} \rightarrow \infty$. Let H_m be an orthogonal copy of E_{N_m} , write $(v_{m,j})_{j=1}^{N_m}$ for the copied basis, and retain the notation

$$S_m : \ell_2^{N_m} \longrightarrow H_m, \quad S_m e_j = v_{m,j}.$$

Set

$$H = \left(\bigoplus_{m=1}^{\infty} H_m \right)_{\ell_2}$$

and order the vectors $(v_{m,j})$ block by block.

Lemma 2. *The blockwise concatenation $(v_{m,j})$ is a normalized Schauder basis of H with basis constant at most $\max(1, K)$, and it is not a Riesz basis.*

Proof. Let $P_{m,r}$ be the projection in H_m onto the first r copied basis vectors. It is the restriction of the corresponding partial-sum projection of (u_j) , so $\|P_{m,r}\| \leq K$. A global partial-sum projection retains all blocks before some m , applies $P_{m,r}$ in block m , and annihilates later blocks. Orthogonality gives, for $x = (x_q)_q$,

$$\|(x_1, \dots, x_{m-1}, P_{m,r}x_m, 0, \dots)\|^2 \leq \max(1, K^2) \sum_q \|x_q\|^2.$$

Uniform boundedness of these projections and density of the finite block vectors prove the Schauder basis assertion.

If the concatenation had Riesz bounds independent of the coefficients, their restriction to block m would uniformly bound both $\|S_m\|$ and $\|S_m^{-1}\|$. This would bound κ_m , a contradiction. $\square$

We now assign frequencies. For $m \geq 1$ put

$$a_m = 4^m, \quad \delta_m = \frac{2^{-m}}{\kappa_m N_m}, \quad \beta_{m,j} = a_m + \delta_m j \quad (1 \leq j \leq N_m).$$

Since $0 < \delta_m N_m \leq 2^{-m} < 1$, the frequencies in block m lie in $(4^m, 4^m + 1)$. Thus all $\beta_{m,j}$ are positive and pairwise distinct. On H_m define

$$T_m(t) = S_m \operatorname{diag}(e^{i\beta_{m,1}t}, \dots, e^{i\beta_{m,N_m}t}) S_m^{-1}.$$

After removing the common scalar phase e^{ia_mt} ,

$$\begin{aligned} \|T_m(t)\| &\leq 1 + \|S_m\| \|S_m^{-1}\| \max_{1 \leq j \leq N_m} |e^{i\delta_m j t} - 1| \\ &\leq 1 + \kappa_m \delta_m N_m |t| = 1 + 2^{-m} |t|. \end{aligned} \tag{1}$$

Here we used $|e^{is} - 1| \leq |s|$ for real s .

Define the orthogonal direct sum

$$T(t) = \bigoplus_{m=1}^{\infty} T_m(t).$$

Estimate (1) shows that this is a bounded operator and

$$\|T(t)\| = \sup_m \|T_m(t)\| \leq 1 + \frac{|t|}{2}. \tag{2}$$

The group law holds in every block and hence globally. On the dense subspace of vectors supported in finitely many blocks, $T(t)x \rightarrow x$ as $t \rightarrow 0$ by finite-dimensional continuity. Estimate (2) is uniform for $|t| \leq 1$, so a density argument extends this convergence to every $x \in H$. Thus $(T(t))_{t \in \mathbb{R}}$ is a C_0 -group.

For completeness, we identify its generator without invoking a direct-sum generator theorem. Put

$$A_m = S_m \operatorname{diag}(i\beta_{m,1}, \dots, i\beta_{m,N_m}) S_m^{-1}$$

and define

$$\begin{aligned} D(A) &= \{x = (x_m)_m \in H : (A_m x_m)_m \in H\}, \\ Ax &= (A_m x_m)_m. \end{aligned} \tag{3}$$

If $x \in D(A)$ and $y = Ax$, the finite-dimensional identities

$$T_m(t)x_m - x_m = \int_0^t T_m(s)A_m x_m ds$$

assemble, using strong continuity and the local bound, into

$$T(t)x - x = \int_0^t T(s)y ds.$$

Dividing by t and letting $t \rightarrow 0$ proves that x belongs to the generator domain and that the generator sends x to Ax . Conversely, if x belongs to the generator domain, applying the bounded orthogonal projection onto H_m to its difference quotient gives $(Ax)_m = A_m x_m$. The generator value lies in H , so $(A_m x_m)_m \in H$. This proves (3) is exactly the generator domain.

Finally,

$$Av_{m,j} = i\beta_{m,j}v_{m,j}.$$

The eigenvalues are distinct and purely imaginary, and Lemma 2 gives the required normalized non-Riesz Schauder basis of eigenvectors. Moreover,

$$\|Av_{m,1}\| = \beta_{m,1} \geq 4^m$$

while $\|v_{m,1}\| = 1$, so A is unbounded. This proves the theorem. $\square$

5 Verification report

The proof is analytic and does not depend on finite computation. The checks most likely to expose a defect were carried out separately:

1. Uniform partial-sum bounds survive concatenation because the blocks are orthogonal; Lemma 2 gives the exact estimate.
2. The basis cannot accidentally become Riesz, because Riesz bounds would hold on every block and contradict $\kappa_m \rightarrow \infty$.
3. The only amplification by the ill-conditioned basis is $\kappa_m \delta_m N_m$, which equals 2^{-m} exactly. This proves boundedness for every t and supplies the uniform local bound needed for strong continuity.
4. Formula (3) was verified in both directions through the integral identity and block projections. Unboundedness is witnessed on unit eigenvectors, not inferred merely from the written eigenvalue formula.

No unproved lemma or computational dependency remains. A longer checklist is included as `verification_report.md` in the packet.

6 Novelty check, limitations, and review recommendation

A bounded search was performed on 9 August 2026 in the run’s local arXiv corpus and through web/arXiv search. Queries included the exact sentence “unbounded generator ... bounded non-Riesz basis”, the authors’ names, and combinations of “conditional Schauder basis”, “eigenvectors”, and “ C_0 -group”. The search also inspected the authors’ later arXiv:1809.03079 and the related 2021 work on resolvents and polynomial bounds. Those works treat complete minimal eigenvector families that do not form a Schauder basis; no later source explicitly answering the bounded Schauder non-Riesz question was found. This is a bounded novelty check, not a certification of priority.

The theorem answers the construction question but does not address Conjecture 20 of the source paper, which imposes a finite-union uniform-gap condition on the spectrum. Indeed, the close frequency clusters here deliberately violate that condition. Human review should focus on Lemma 2 and estimate (1); the rest is a standard but fully written direct-sum group argument.

References

- [1] G. M. Sklyar and V. Marchenko, *Hardy inequality and the construction of infinitesimal operators with non-basis family of eigenvectors*, J. Funct. Anal. **272** (2017), 1017–1043; arXiv:1405.2731v4.
- [2] K. I. Babenko, *On conjugate functions*, Dokl. Akad. Nauk **62** (1948), no. 2, 157–160 (Russian), MR0027093.